

Detection Rules Report

Milestone 3: Detection & Visualization

1. Introduction

During Milestone 2, AI-assisted network traffic analysis was performed on the PCAP-derived CSV datasets to identify suspicious communication patterns, unusual traffic behaviour, and potential security concerns.

The analysis identified several network indicators requiring monitoring:

- Repeated communication between internal and external hosts
- High-volume communication with specific external IP addresses
- Use of unencrypted HTTP communication
- Large data exchange patterns
- DNS-related traffic requiring additional validation

Based on these observations, detection rules were developed to help a SOC analyst identify similar network behaviours in future investigations. These rules are designed to identify suspicious patterns and generate investigation alerts; they do not automatically confirm malicious activity.

2. Detection Rule Development Methodology

The detection rules were created using the following approach:

1. Identify repeatable network behaviour from Milestone 2 findings
2. Define conditions that can be monitored automatically
3. Generate alerts when suspicious thresholds are reached
4. Consider possible legitimate explanations to reduce false positives
5. Validate alerts through additional security investigation

Each rule follows the same basic detection structure: when a defined condition occurs, the system generates a security alert.

```
IF condition occurs
  THEN generate security alert
```

3. Detection Rule 1: Repeated External Communication Detection

Rule Name

Possible Automated External Communication Detection

Detection Logic

```
IF:
  An internal host communicates repeatedly with the same external IP address
```

```
AND communication count exceeds a defined threshold
AND connections occur within a short time period
```

Example:

```
Internal Host:      10.10.0.33
External Destination: 18.184.26.60
Pattern:           High packet exchange volume
```

THEN:

```
ALERT: "Possible automated external communication detected."
```

Security Purpose

Repeated communication with a single external destination may indicate automated applications, background services, software updates, or, in some cases, command-and-control style behaviour. This rule helps identify systems that require further investigation.

Detection Capability Against Investigated Traffic

This rule would identify the communication pattern observed between 10.10.0.33 and 18.184.26.60, where a high number of packets were exchanged.

False Positive Possibilities

Possible legitimate causes include web applications, cloud services, software updates, and browser activity. Additional validation is required before classification.

4. Detection Rule 2: High Volume HTTP Communication Detection

Rule Name

Unencrypted HTTP Communication Monitoring

Detection Logic

IF:

```
An internal host communicates with an external destination
AND destination port is TCP/80
AND communication volume exceeds normal baseline
```

Example:

```
Source:      10.10.0.33
Destination: 18.184.26.60
Protocol:    TCP
Port:       80
```

THEN:

```
ALERT: "High-volume unencrypted HTTP communication detected."
```

Security Purpose

HTTP traffic is not encrypted and may expose web requests, application information, and sensitive data. Although HTTP is commonly used legitimately, abnormal HTTP activity should still be reviewed.

Detection Capability Against Investigated Traffic

This rule would detect the HTTP communication observed in Segment 2, from 10.10.0.33 to 18.184.26.60 on port 80.

False Positive Possibilities

Possible legitimate activity includes website browsing, legacy applications, and public web services.

5. Detection Rule 3: Large Data Transfer Detection

Rule Name

Unusual Data Transfer Volume Detection

Detection Logic

```
IF:
  An internal host transfers unusually large amounts of data
  AND the destination is external
  AND traffic volume exceeds normal user behaviour

THEN:
  ALERT: "Unusual external data transfer detected."
```

Security Purpose

Large transfers may reflect ordinary activity such as file downloads, application activity, or software updates. They may also warrant investigation for unauthorized transfer, data leakage, or abnormal application behaviour.

Detection Capability Against Investigated Traffic

This rule relates to the Segment 3 observation of traffic from 10.10.0.33 to 8.8.8.8, where the largest observed frame reached 58,337 bytes.

False Positive Possibilities

Possible legitimate causes include DNS or application responses, cloud services, and normal application traffic.

6. Detection Rule 4: Suspicious DNS Activity Detection

Rule Name

DNS Traffic Anomaly Detection

Detection Logic

```
IF:
  DNS traffic volume increases unexpectedly
  AND DNS responses contain unusually large data
  AND queries show abnormal patterns
  (e.g. long domain names, high query frequency, unusual DNS record types)

THEN:
  ALERT: "Possible DNS anomaly detected."
```

Security Purpose

DNS is normally used for domain resolution, but attackers may abuse it for command communication, DNS tunnelling, or data transfer.

Detection Capability Against Investigated Traffic

This rule relates to Segment 3 findings involving communication between 10.10.0.33 and 8.8.8.8, and the large DNS-related frame observations noted there.

False Positive Possibilities

Legitimate causes include DNS caching, application lookups, and cloud service resolution.

7. Detection Rule 5: Unknown External Destination Communication Detection

Rule Name

New External Communication Monitoring

Detection Logic

```
IF:
  Internal host communicates with a previously unseen external IP address
  AND the destination is not present in approved communication lists
  AND communication frequency increases

THEN:
  ALERT: "New external communication requires investigation."
```

Security Purpose

Unexpected external connections may indicate new applications, misconfiguration, or suspicious software activity.

Detection Capability Against Investigated Traffic

This rule would help monitor external destinations identified during analysis, including 18.184.26.60, 204.79.197.200, 204.11.56.48, and 34.102.136.180.

False Positive Possibilities

Legitimate examples include new websites, cloud infrastructure, and software updates.

8. False Positive Considerations

Detection rules may generate alerts from legitimate network activity. Common false positive sources include web browsing, cloud applications, software updates, antivirus communication, and operating system services.

Alerts should therefore be investigated using endpoint logs, application information, user activity records, threat intelligence, and packet inspection.

9. Validation Against Milestone 2 Findings

The detection rules were derived from confirmed observations identified during AI-assisted analysis.

Milestone 2 Observation	Detection Rule Applied
Repeated communication with external IP	Rule 1
High-volume HTTP traffic	Rule 2
Large data exchange patterns	Rule 3
DNS-related unusual traffic	Rule 4
Multiple external destinations	Rule 5

10. Conclusion

The developed detection rules provide a foundation for identifying suspicious network behaviour observed during the investigation. These rules focus on behavioural indicators rather than automatically classifying traffic as malicious.

Future validation should combine network monitoring, endpoint analysis, threat intelligence, and security logging. The rules can assist SOC analysts by highlighting potentially abnormal activity requiring further investigation.